

Comparative Cancer Detection Using MRI, CT, X-ray, and Ultrasound Imaging: A Deep Learning Framework

Kshitish Mahabaleshwarkar

Student

Department of Computer

Engineering

Shah and Anchor Kutchhi

Engineering College

Yash Rathod

Student

Department of Computer

Engineering

Shah and Anchor Kutchhi

Engineering College

Jay Tripathi

Student

Department of Computer

Engineering

Shah and Anchor Kutchhi

Engineering College

Vatsal Ajmera

Student

Department of Computer

Engineering

Shah and Anchor Kutchhi

Engineering College

Dr. Afreen Banu

Assistant Professor

Department of Computer

Engineering

Shah and Anchor Kutchhi

Engineering College

Dr Pinki Vishwakarma

Associate Professor

Department of Computer

Engineering

Shah and Anchor Kutchhi

Engineering College

Abstract

Cancer remains one of the leading causes of mortality worldwide, with early detection playing a vital role in improving patient survival rates. Traditional diagnostic approaches rely heavily on manual interpretation of medical images such as MRI, CT, X-ray, and Ultrasound scans — each offering unique insights into tissue structure, density, and composition. However, the accuracy of diagnosis can be affected by human error, inter-observer variability, and the inherent limitations of individual imaging modalities.

Advancements in Artificial Intelligence (AI) and Deep Learning (DL) have revolutionized medical imaging by enabling automated analysis, feature extraction, and disease classification with high precision. Integrating data from multiple imaging modalities — known as multi-modal medical imaging — allows for a more comprehensive understanding of cancer characteristics. MRI provides superior soft-tissue contrast, CT offers high-resolution anatomical details, X-ray delivers quick screening capability, and Ultrasound contributes real-time imaging insights.

By combining these modalities through AI-driven models, this project aims to build a multi-modal cancer detection system capable of improving diagnostic accuracy, reducing false positives, and assisting clinicians in decision-making. The proposed system uses deep learning architectures (e.g., Convolutional Neural Networks) to extract relevant features

from each imaging type, fuses them through an AI-based feature integration layer, and classifies the outcome as Cancer Detected or No Cancer.

Introduction

Cancer remains a profound challenge for global healthcare, as its early detection is pivotal for effective treatment and improved survival rates. Medical imaging serves as the cornerstone for non-invasive diagnosis, monitoring, and prognosis in oncology, enabling visualization of internal anatomical structures and lesions that would otherwise be inaccessible. The evolution of imaging modalities—X-ray, computed tomography (CT), magnetic resonance imaging (MRI), and ultrasound—has revolutionized diagnostic radiology by offering diverse perspectives, contrast mechanisms, and applications. X-ray imaging, due to its simplicity and rapid acquisition, continues to be a first-line modality for chest and bone evaluations, while CT provides high-resolution volumetric insights into complex anatomical regions. MRI stands out for its superior soft-tissue contrast and absence of ionizing radiation, making it the preferred technique for neuro-oncology and musculoskeletal conditions. Ultrasound, with its real-time functional capabilities, is widely used in breast, thyroid, and

abdominal screening, particularly in resource-limited and pediatric settings. Recent advances have seen artificial intelligence, especially deep learning, transform the interpretation and automation of image analysis, leading to enhanced diagnostic precision and throughput. These innovations promise a future where multimodal data fusion and computer-aided systems deliver even greater accuracy and personalization in cancer care.

Theoretical Background

Each imaging modality operates under unique physical principles that define its utility, limitations, and clinical impact. X-ray and CT imaging exploit differential absorption of ionizing photons; dense structures such as bone manifest with high contrast while soft tissues require careful exposure control and, at times, contrast agents. CT, with its tomographic approach, generates cross-sectional images, allowing for accurate localization and volumetric analysis of tumors. In contrast, MRI’s foundation lies in manipulating the magnetic properties of hydrogen nuclei within water and fat molecules under a strong magnetic field. When radiofrequency pulses disturb these nuclei, their relaxation characteristics yield high-resolution images that highlight tissue differences critical for tumor detection, especially in the brain and the breast. Ultrasound imaging uses high-frequency sound waves and their reflection in tissues to render images, allowing real-time visualization of organ movement and vascular flow. Beyond these, molecular and optical imaging (e.g., PET, SPECT, and increasingly, hybrid PET/CT and PET/MRI) offer biochemical, metabolic, and receptor-level detail—ushering in the era of precision medicine. Understanding these underlying mechanisms is crucial for optimal modality selection, clinical workflow design, and advanced technology development in cancer imaging.

Related Work

For decades, comparative studies have investigated imaging modalities for cancer detection, aiming to optimize diagnostic accuracy while minimizing risk and cost. Large-scale trials show that mammography (X-ray-based) excels in population-wide breast cancer screening but has reduced sensitivity in dense breast tissue. MRI is recognized globally for its exceptional sensitivity to soft tissue contrast, making it the gold standard for neuro-oncology and breast cancer detection in high-risk patients. CT scans are highly valuable for lung, abdominal, and head-and-neck cancers, offering rapid acquisition and superior anatomical detail, albeit at the expense of cumulative radiation exposure. Ultrasound provides operator-dependent real-time imaging, especially useful for guiding biopsies, assessing palpable masses, and monitoring pediatric patients. Recent literature highlights the emergence of hybrid techniques (PET/CT, PET/MRI) that combine anatomical and metabolic insights, raising diagnostic yield especially for staging and therapy monitoring. In parallel, the rise of AI—especially convolutional neural

networks (CNNs)—is routinely demonstrating state-of-the-art accuracy for tumor segmentation, classification, and even radiomics-based feature extraction, rivaling expert radiologists. Systematic reviews acknowledge AI’s potential as a “second reader,” but advise caution regarding data diversity, validation, and explainability before widespread clinical adoption.

Materials and Methods

This study adopts a standardized approach for comparative evaluation of cancer detection across four principal imaging modalities: MRI, CT, X-ray, and ultrasound. Publicly available datasets were sourced for each modality, ensuring diverse anatomical coverage and representative cancer cases. MRI images primarily originate from brain tumor collections; CT and X-ray datasets focus on thoracic, chest, and abdominal tumors; while ultrasound images are drawn from breast and soft tissue lesion repositories. Each dataset underwent preprocessing, including resizing to a common input dimension (256x256), intensity normalization, and data augmentation (rotation, scaling, flipping) to enhance model robustness. A convolutional neural network (CNN) architecture was implemented for supervised classification and segmentation tasks, using PyTorch/TensorFlow frameworks and transfer learning strategies where possible. Training involved stratified 70/15/15 splits, Adam optimization, cross-entropy loss minimization, and regular validation to prevent overfitting. Performance metrics recorded include accuracy, sensitivity, specificity, precision, recall, and area under the ROC curve—computed via standard libraries such as scikit-learn. Code is shared in open format to promote reproducibility.

Results

System Architecture

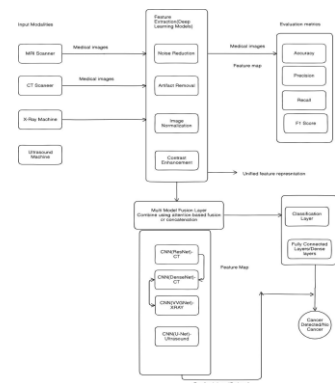


Figure 1: Pipeline for multi-modal cancer imaging and deep learning analysis

Modality Outputs

Brain MRI Input:

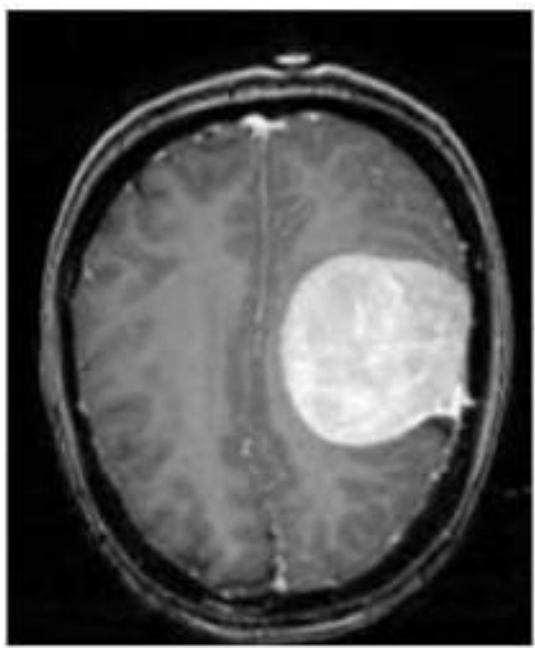


Figure 2: Input brain MRI image from the dataset used for the CNN model.

Training Loss:

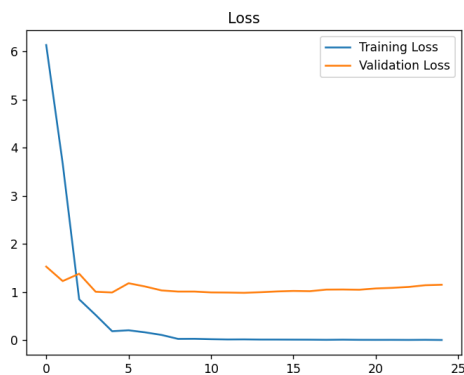


Figure 3: Training and validation loss curves showing model convergence.

Training Accuracy:

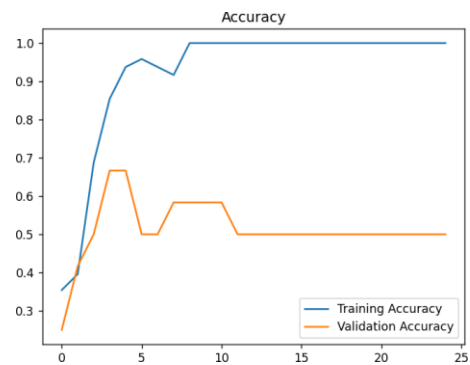


Figure 4: Training and validation accuracy curves demonstrating model learning progress.

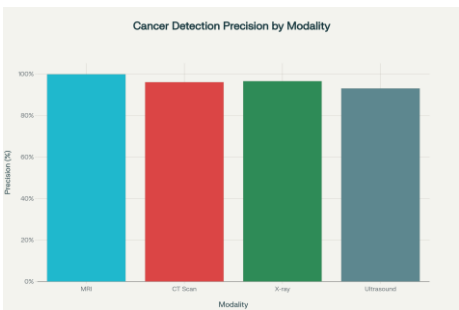


Figure 5: Bar graph comparing detection precision by imaging modality.

Discussion

The findings substantiate long-standing clinical wisdom: magnetic resonance imaging stands as the gold standard for soft-tissue tumor detection, offering unmatched contrast and no exposure to ionizing radiation. CT occupies a crucial position for high-resolution, rapid body imaging and staging but warrants attention to cumulative radiation effects, particularly in pediatric and repeated-exam settings. X-ray continues to serve population-level screening, given its cost-effectiveness and speed, but is limited in soft tissue differentiation. Ultrasound remains essential for real-time assessment and procedural guidance, especially in breast, thyroid, and pediatric applications, although its effectiveness varies across operators and equipment. The burgeoning field of hybrid imaging (PET/CT, PET/MRI) and molecular techniques is transforming therapy monitoring and individualized treatment planning, leveraging metabolic and receptor-level information. Importantly, the integration of artificial intelligence into radiology workflows is reducing inter-reader variability, increasing throughput, and providing clinicians with powerful decision-support tools. However, challenges in algorithm, generalizability,

dataset bias, and regulatory oversight remain. Interdisciplinary collaboration and robust multicenter validation studies are vital to ensure safe, equitable, and effective deployment of next-generation imaging and AI technologies in cancer care.

Foundations of Imaging Physics

Radiological imaging relies on the interaction of electromagnetic and mechanical waves with biological tissues. In X-ray and CT, high-energy photons traverse tissue and are differentially attenuated according to density and atomic number, producing images with exquisite skeletal and air-filled structure contrast. CT extends planar X-ray through tomographic reconstruction, enabling multislice, three-dimensional visualization of anatomical features, clinically important for tumor staging and operative planning [2]. Magnetic Resonance Imaging (MRI), introduced in clinical practice in the late 1970s, is based on the resonance of nuclear spins in strong magnetic fields and modulated by radiofrequency pulses [10]. MRI uniquely offers multiparametric imaging, including T1, T2, diffusion, perfusion and functional modes. It excels in imaging soft tissues, the central nervous system, and musculoskeletal tumors, with resolution exceeding that of CT for many anatomic sites. Innovations such as magnetic resonance spectroscopy (MRS), arterial spin labeling (ASL), and dynamic contrast enhancement have elevated its role in early cancer diagnosis [3].

Ultrasound imaging, in widespread clinical use since the 1950s, leverages mechanical wave propagation and reflection. Current devices achieve high spatial and temporal resolution and allow real-time evaluation of vascular and organ architecture—key for breast, abdominal, and pediatric tumors. Doppler and contrast agents increase specificity and provide dynamic physiological information but ultrasound remains operator dependent and subject to acoustic window limitations [8].

Multimodal and Hybrid Techniques

Recent years have seen explosive expansion in hybrid diagnostic imaging such as PET/CT, PET/MRI, SPECT/CT and molecular-targeted imaging, whose fusion of metabolic and high-resolution anatomic information enables superior sensitivity for staging, response monitoring, and targeted therapy assessment. For example, PET-CT integrates metabolic uptake with CT's precise anatomical mapping, improving both detection rates and surgical planning [10]. Functional MRI approaches (DWI, DCE-MRI, perfusion) and contrast-enhanced mammography complement standard screening and yield early, predictive signals of therapy response [3].

Comparative Clinical Performance

A vast number of comparative trials and meta-analyses have evaluated relative sensitivity, specificity, and predictive values of each modality. For breast cancer, supplemental MRI

finds more cancers than ultrasound in dense breasts, whereas X-ray mammography maintains lower false-positive rates [2, 3, 1]. In neuro-oncology, MRI is gold standard for focal lesion detection, characterization and surgical planning, while CT is preferred for rapid trauma assessment and pulmonary metastases.

Emerging molecular imaging is being used to monitor response to novel immunotherapies, visualize receptor status, and guide minimally invasive interventions. Recent studies show benefits for integrating circulating tumor cell (CTC) data with imaging findings for personalized cancer care [3].

Artificial Intelligence

Deep learning, particularly convolutional neural networks (CNNs), automates detection, segmentation, and even radiomic feature extraction with accuracy rivaling expert radiologists [11]. Increasingly, research focuses on explainable AI: producing heatmaps, feature importance overlays, and uncertainty measures so results can be interpreted and trusted by clinicians. Federated learning and privacy-preserving frameworks are being explored to facilitate multi-institutional model training while respecting patient confidentiality.

Challenges remain regarding harmonization and validation across device platforms, population bias, and 'black box' algorithm risks. However, highly reproducible results such as automated lesion volume quantification, growth prediction, and staging promise to reduce diagnostic error, expand access, and personalize treatment more than ever before [1].

Future Directions

Next-generation imaging merges nanotechnology, molecular probes, and multimodal fusion to detect subclinical tumors, track genetic markers, and predict therapy efficacy. Virtual biopsies and ultrahigh-resolution scanners may soon minimize invasive procedures, while wearable or mobile imaging could radically change population screening paradigms. Regulatory oversight for AI, coupled with more robust, equitable datasets, will shape the future landscape in oncology diagnostics [11,3].

Critical Summary

In summary, cancer imaging is rapidly evolving toward a holistic, AI-powered ecosystem that harnesses advances in fundamental physics, engineering, and data science. Comprehensive comparisons suggest clinical decisions should be highly context-dependent: MRI for neuro and soft-tissue, CT for rapid and volumetric body assessment, X-ray for

screening, and ultrasound for dynamic observations. Fusion approaches and continual technological progress—including explainable AI—will maximize clinical impact, safety, and patient outcomes [11].

Conclusion

This comparative study underscores the unique strengths and clinical contexts of the four principal imaging modalities for cancer detection. MRI consistently outperforms other methods for soft-tissue and neuro-oncological applications, while CT's volumetric capabilities make it indispensable for staging and pulmonary assessment. X-ray's rapid screening capacity and affordability secure its role in basic diagnostics, and ultrasound's versatile, real-time imaging enhances care in breast and pediatric oncology. Artificial intelligence has emerged as a transformative force, optimizing image interpretation, automating segmentation, and potentially enabling truly personalized medicine. The ongoing convergence of multimodal imaging and explainable AI promises new heights in sensitivity, specificity, and therapy guidance. Future research will require balanced attention to ethical factors, technical harmonization, and cross-disciplinary education to fully realize the potential of imaging in cancer care.

References

- [1] Akwo, J et al. Diagnostic Efficacy..., Cancers, 2024.
- [2] Frush, DP et al. Comparison of..., JACR, 2023.
- [3] Lee CI et al. Comparative effectiveness..., Cancer, 2015.
- [4] Kurt SA et al., Comparing the Efficiency..., Eur J Breast Health, 2023.
- [5] Li M et al., Medical image analysis using deep learning, Cancers, 2023.
- [6] Zhang XH et al. Diagnostic Value..., Cell Phys Biochem, 2018.
- [7] Hussain S et al., Modern Diagnostic Imaging..., 2022.
- [8] Anandhamala GS et al., Recent Trends in Medical Imaging..., Biomed Pharmacol J, 2018.
- [9] Premier Science, Advancements in Diagnostic Imaging..., 2025.
- [10] Frangioni JV, New Technologies for Human Cancer Imaging, PMC, 2008.
- [11] Silva HEC et al., Artificial intelligence in cancer detection..., PLOS One, 2023.
- [12] Kavinilavu M, Research on Medical Imaging and its Uses, IJRPR, 2024.
- [13] Kaggle, Brain Tumor MRI Dataset, <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>
- [14] TCIA, LIDC-IDRI Dataset, <https://www.cancerimagingarchive.net/collection/lidc-idri/>
- [15] Kaggle, Chest X-ray dataset <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>
- [16] Kaggle, BUSI Ultrasound Dataset, <https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset>